The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 January 2013.

INCH-POUND

MIL–PRF–19500/734A
16 October 2012
SUPERSEDING
MIL–PRF–19500/734
16 February 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER DARLINGTON
TYPES 2N7569, 2N7570, AND 2N7571, JAN, JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL–PRF–19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, power darlington transistors. Four levels of product assurance are provided for each device type as specified in MIL–PRF–19500.

1.2 Physical dimensions. The device package style is TO–204AA (formerly TO–3) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T		$R_{\theta JC}$ (2)	V_{CBO} and V_{CEX}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}
	T_A = +25°C	T_C = +25°C (1)							
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	°C
2N7569	6	175	1.0	350	300	4.0	2.5	20	–65 to +200
2N7570	6	175	1.0	400	350	4.0	2.5	20	–65 to +200
2N7571	6	175	1.0	450	400	4.0	2.5	20	–65 to +200

(1) See figure 2 for temperature-power derating curves.
(2) See figure 3 (thermal impedance graph).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime ATTN: VAC, P.O. Box 3990, Columbus, OH 43218–3990, or emailed to semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Limits	h_{FE1} (1) V_{CE} = 5 V dc I_C = 10 A dc	h_{FE2} (1) V_{CE} = 5V dc I_C = 20 A dc	$V_{BE(sat)}$ I_C = 15 A dc I_B = 1.2 A dc	$V_{CE(sat)1}$ I_C = 15 A dc I_B = 1.2 A dc	$V_{CE(sat)2}$ I_C = 20 A dc I_B = 1.2 A dc
			V dc	V dc	V dc
Min	75	50			
Max	500		2.6	1.5	1.7

Limits	$\|h_{fe}\|$	C_{obo}	Pulse response (2)			
	V_{CE} = 10 V dc I_C = 1 A dc f = 1 MHz	V_{CB} = 10 V dc I_E = 0 100 KHz < f < 1 MHz	t_d	t_r	t_s	t_f
		pF	µs	µs	µs	µs
Min	25	150				
Max		375	0.1	0.3	1.2	0.3

(1) Pulsed (see 4.5.1).
(2) See figure 4 for pulse response circuits.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL–PRF–19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL–STD–750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://assist.dla.mil/quicksearch or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

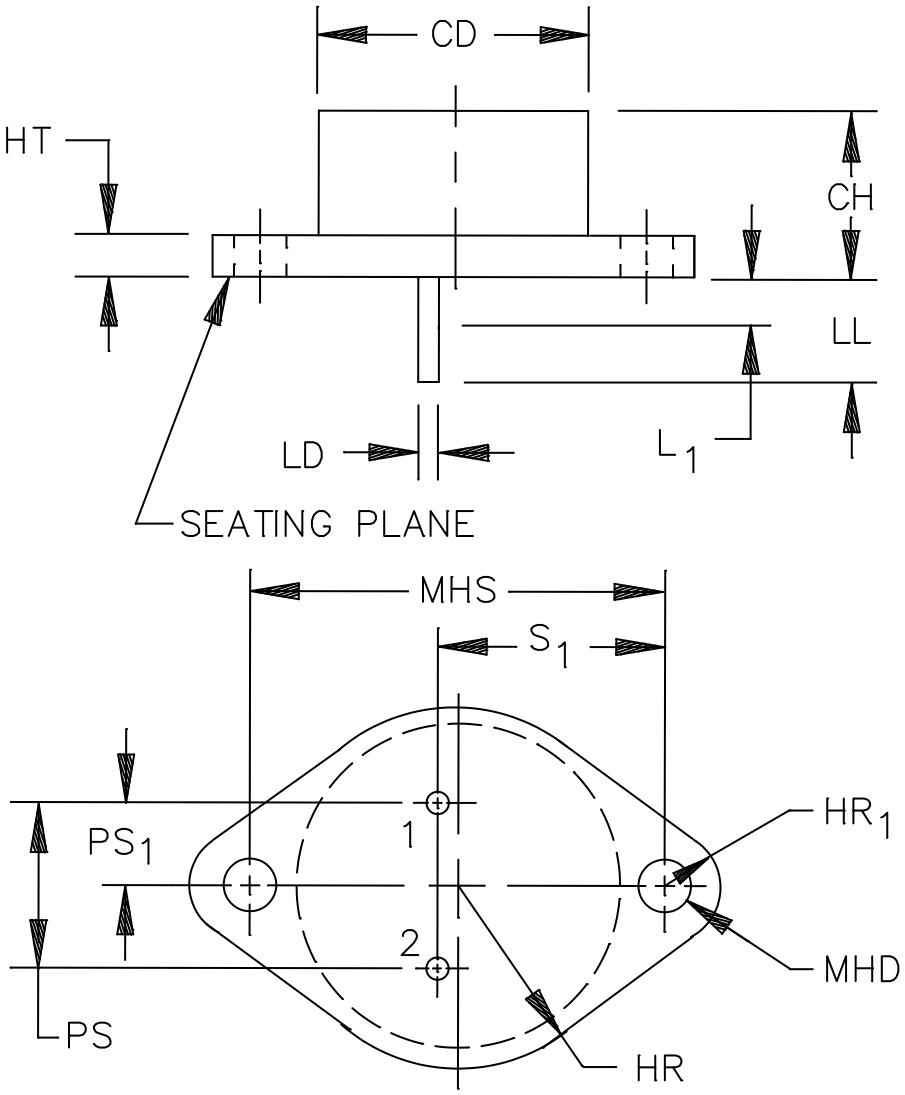


FIGURE 1. Physical dimensions (TO–204AA, formerly TO–3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.23	2
CH	.250	.450	6.35	11.43	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	3
HT	.060	.135	1.52	3.43	
L_1		.050		1.27	4, 5
LD	.038	.043	0.97	1.09	4, 5
LL	.312	.500	7.92	12.70	4
MHD	.151	.161	3.84	4.09	6
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	4, 7, 8
PS_1	.205	.225	5.21	5.72	4, 7, 8
S_1	.655	.675	16.64	17.15	7

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Body contour is optional within zone defined by dimension CD.
3. At both ends.
4. Both terminals.
5. Dimension LD applies between L_1 and LL. Lead diameter shall not exceed twice dimension LD within dimension L_1. Diameter is uncontrolled in dimension L_1.
6. Two holes.
7. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below the seating plane. When gauge is not used, measurement shall be made at seating plane.
8. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
9. Terminal 1 is the emitter, terminal 2 is base. The collector shall be electrically connected to the case.
10. In accordance with ASME Y14.5M, diameters are equivalent to ϕ symbology.

FIGURE 1. Physical dimensions (TO–204AA, formerly TO–3) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL–PRF–19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL–PRF–19500 and as follows:

I_H – The collector current applied to the device under test during the heating period.
I_M – The measurement current applied to forward bias the junction for measurement of V_{BE}.
t_H – The duration of the applied heating power pulse.
t_{SW} – Sample window time during which final V_{BE} measurement is made.

3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL–PRF–19500 and on figure 1 (TO–204AA, similar to TO–3) herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL–STD–750, MIL–PRF–19500, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Polarity. The identification of terminals of the device package shall be as shown on figure 1. Terminal 1 shall be connected to the emitter and terminal 2 shall be connected to the base. The collector shall be electrically connected to the case.

3.5 Marking. Marking shall be in accordance with MIL–PRF–19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL–PRF–19500, and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with table E–IV of MIL–PRF–19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E–IV of MIL–PRF–19500)	Measurement	
	JANS	JANTX and JANTXV
3c (1)	Thermal impedance (see 4.3.1)	Thermal impedance (see 4.3.1)
7	Optional.	Optional.
9	I_{CEX1}	I_{CEX1}
11	I_{CEX1} and h_{FE2}, ΔI_{CEX1} = 100 percent of initial value or 100 nA dc, whichever is greater.	I_{CEX1} and h_{FE2}, ΔI_{CEX1} = 100 percent of initial value or 100 nA dc, whichever is greater.
12	See 4.3.2	See 4.3.2
13	See subgroups 2 and 3 of table I herein. ΔI_{CEX1} = 100 percent of initial value or 50 nA dc, whichever is greater. Δh_{FE2} = ±15 percent of initial value.	See subgroup 2 of table I herein. ΔI_{CEX1} = 100 percent of initial value or 100 nA dc, whichever is greater. Δh_{FE2} = ±25 percent of initial value.
14	Required.	Required.

(1) This test shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with test method 3131 of MIL–STD–750 using the guidelines in that test method for determining I_H, I_M, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs maximum. The thermal impedance limit used in 4.3, screen 3c, and table I, subgroup 2 herein shall be set statistically by the supplier. See group E inspection (table II, subgroup 4) herein.

4.3.2 Power burn-in conditions. Power burn-in conditions shall be as follows: T_J = +175°C minimum, V_{CB} greater than or equal to 25 V dc; T_A = +30°C maximum.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL–PRF–19500, and as specified herein. If alternate screening is being performed in accordance with MIL–PRF–19500, a sample of screened devices shall be submitted to and pass the requirements of table I, subgroup 1 and 2 inspection only. Group B inspection in accordance with subgroup 1 of table E–VIB of MIL–PRF–19500 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E–V of MIL–PRF–19500 and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E–VIA (JANS) or table E–VIB (JAN, JANTX, and JANTXV) of MIL–PRF–19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with the applicable step of 4.6 herein.

4.4.2.1 Product assurance level JANS (table E–VIA of MIL–PRF–19500).

Subgroup	Method	Conditions
B4	1037	For solder die attach: 2,000 cycles, V_{CB} greater than or equal to 100 V dc.
B5	1027	For eutectic die attach: V_{CB} greater than or equal to 100 V dc; adjust P_T to achieve T_J = +175°C minimum; T_A = +30°C maximum

4.4.2.2 Product assurance levels JAN, JANTX, and JANTXV (table E–VIB of MIL–PRF–19500).

Subgroup	Method	Conditions
B3	1027	For eutectic die attach: V_{CB} greater than or equal to 25 V dc; adjust P_T to achieve T_J = +175°C minimum; T_A = +30°C maximum
B3	1037	For solder die attach: 2,000 cycles, V_{CB} greater than or equal to 100 V dc.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E–VII of MIL–PRF–19500 and as follows herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with the applicable step of table II herein.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 10 pounds; time = 15 s.
C5	3131	See 4.3.1, $R_{\theta JC}$ = 1.0°C/W.
C6	1027	For eutectic die attach: V_{CB} greater than or equal to 100 V dc; adjust P_T to achieve T_J = +175°C minimum; T_A = +30°C maximum, 1,000 hours minimum.
C6	1037	For solder die attach: 6,000 cycles, V_{CB} greater than or equal to 100 V dc.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in appendix E, table E–IX of MIL–PRF–19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with the applicable step of 4.6 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse response measurements. The conditions for pulse response measurement shall be as specified in section 4 of MIL–STD–750.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

4.6 Delta measurements. The requirements for delta measurements for groups B, C, and E shall be as specified below.

Steps	Inspection (1) (2) (3) (4) (5)	MIL–STD–750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to base cutoff current 2N7569 2N7570 2N7571	3036	Bias condition D V_{CB} = 350 V dc V_{CB} = 400 V dc V_{CB} = 450 V dc	ΔI_{CBO}	100 percent of initial value or 100 nA, whichever is greater.		
2	Forward – current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 20 A dc; pulsed (see 4.5.1)	Δh_{FE2}	±25 percent change from initial reading.		
3	Saturation voltage, collector to emitter	3071	I_C = 15 A dc; I_B = 1.2 A dc, pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$	±200 mV change from previously measured value.		

(1) Devices which exceed the group A limits for this test shall not be acceptable.
(2) The delta electrical measurements for group B, product assurance level JANS shall be as follows:
   a. In addition to the measurements specified for subgroup 4 of table E–VIA of MIL–PRF–19500, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 5 of table E–VIA of MIL–PRF–19500, the measurements of steps 1 and 2 of this table shall also be taken.
(3) The delta electrical measurements for group B, product assurance levels JAN, JANTX, and JANTXV shall be as follows:
   a. In addition to the measurements specified for subgroup 3 of table E–VIB of MIL–PRF–19500, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6 of table E–VIB ofMIL–PRF–19500, the measurements of step 2 of this table shall also be taken.
(4) The delta electrical measurements for group C, all product assurance levels, shall be as follows:
   a. In addition to the measurements specified for subgroup 2 of table E–VII of MIL–PRF–19500, the measurements of step 1 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 3 of table E–VII of MIL–PRF–19500, the measurements of step 1 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 6 of table E–VII of MIL–PRF–19500, the measurements of steps 1, 2, and 3 of this table shall also be taken.
(5) The delta measurements for group E, all product assurance levels, shall be as follows: In addition to the measurements specified for subgroups 1 and 2 of table E–IX of MIL–PRF–19500, the measurements of steps 1, 2, and 3 of this table shall also be taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.1	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N7569				300		V dc
2N7570				350		V dc
2N7571				400		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = –1.5 V dc	I_{CEX1}			
2N7569		V_{CE} = 350 V dc			0.1	mA dc
2N7570		V_{CE} = 400 V dc			0.1	mA dc
2N7571		V_{CE} = 450 V dc			0.1	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = –1.5 V dc	I_{CEO}			
2N7569		V_{CE} = 300 V dc			0.1	mA dc
2N7570		V_{CE} = 350 V dc			0.1	mA dc
2N7571		V_{CE} = 400 V dc			0.1	mA dc
Collector to base cutoff current	3036	Bias condition D;	I_{CBO}			
2N7569		V_{CBO} = 350 V dc			0.1	mA dc
2N7570		V_{CBO} = 400 V dc			0.1	mA dc
2N7571		V_{CBO} = 450 V dc			0.1	mA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc	I_{EBO}		300	mA dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 15 A dc; pulsed (see 4.5.1); I_B = 1.2 A dc	$V_{BE(SAT)}$		2.6	V dc
Saturation voltage; collector to emitter	3071	I_C = 15 A dc; pulsed (see 4.5.1) I_B = 1.2 A dc	$V_{CE(SAT)1}$		1.5	V dc
Saturation voltage; collector to emitter	3071	I_C = 20A dc; pulsed (see 4.5.1) I_B = 1.2 A dc	$V_{CE(SAT)2}$		1.7	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE1}	75	500	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 20A dc; pulsed (see 4.5.1)	h_{FE2}	50		

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection – Continued.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = –1.5 V dc	I_{CEX2}			
2N7569		V_{CE} = 350 V dc			5.0	mA dc
2N7570		V_{CE} = 400 V dc			5.0	mA dc
2N7571		V_{CE} = 400 V dc			5.0	mA dc
Low-temperature operation:		T_A = –55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE3}	20		
Subgroup 4						
Magnitude of common emitter small-signal, short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1 A dc; f = 1 MHz	$\|h_{fe}\|$	25		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1.0 MHz	C_{obo}	150	375	pF
Pulse response	3251	Test condition A except test circuit and pulse requirements in accordance with figure 4, T_A = +25°C, V_{CC} = 300V, I_C = 10A I_{B1} = 0.5A, $V_{BE(off)}$ = –5V, tp = 50 μs				
Pulse delay time			t_d		0.1	μs
Pulse rise time			t_r		0.3	μs
Pulse storage time			t_s		1.2	μs
Pulse fall time			t_f		0.3	μs
Subgroup 5		T_C = +25°C				
Safe operating area (continuous dc)	3051	t = 1 s; 1 cycle; see figure 5				
Test 1						
All device types		V_{CE} = 8.0 V dc; I_C = 20 A dc				

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection – Continued.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 – Continued.						
Test 2						
All device types		V_{CE} = 30 V dc; I_C = 3.0 A dc				
Test 3						
All device types		V_{CE} = 100 V dc; I_C = 0.17 A dc				
Test 4						
2N7569		V_{CE} = 300 V dc; I_C = 20 mA dc				
2N7570		V_{CE} = 350 V dc; I_C = 20 mA dc				
2N7571		V_{CE} = 400 V dc; I_C = 20 mA dc				
Safe operating area (switching); inductive collector to emitter sustaining voltage. (clamped at rated V_{CEO})	3053	T_A = +25°C, I_C = 2A, I_{B1} = 0.1A, I_{B2} = 0A, L = 1mH	$V_{CEO(sus)}$			
2N7569		Clamp voltage = 300 V dc		300		V dc
2N7570		Clamp voltage = 350 V dc		350		V dc
2N7571		Clamp voltage = 400 V dc		400		V dc
Safe operating area (switching); inductive collector to emitter sustaining voltage. (clamped at rated V_{CEO})	3053	T_A = +25°C, I_C = 2 A, I_{B1} = 0.5 A, $V_{BE(off)}$ = 1V, L = 1 mH	$V_{CEX(sus)}$			
2N7569		Clamp voltage = 300 V dc		300		V dc
2N7570		Clamp voltage = 350 V dc		350		V dc
2N7571		Clamp voltage = 400 V dc		400		V dc
Subgroups 6 and 7						
Not applicable						

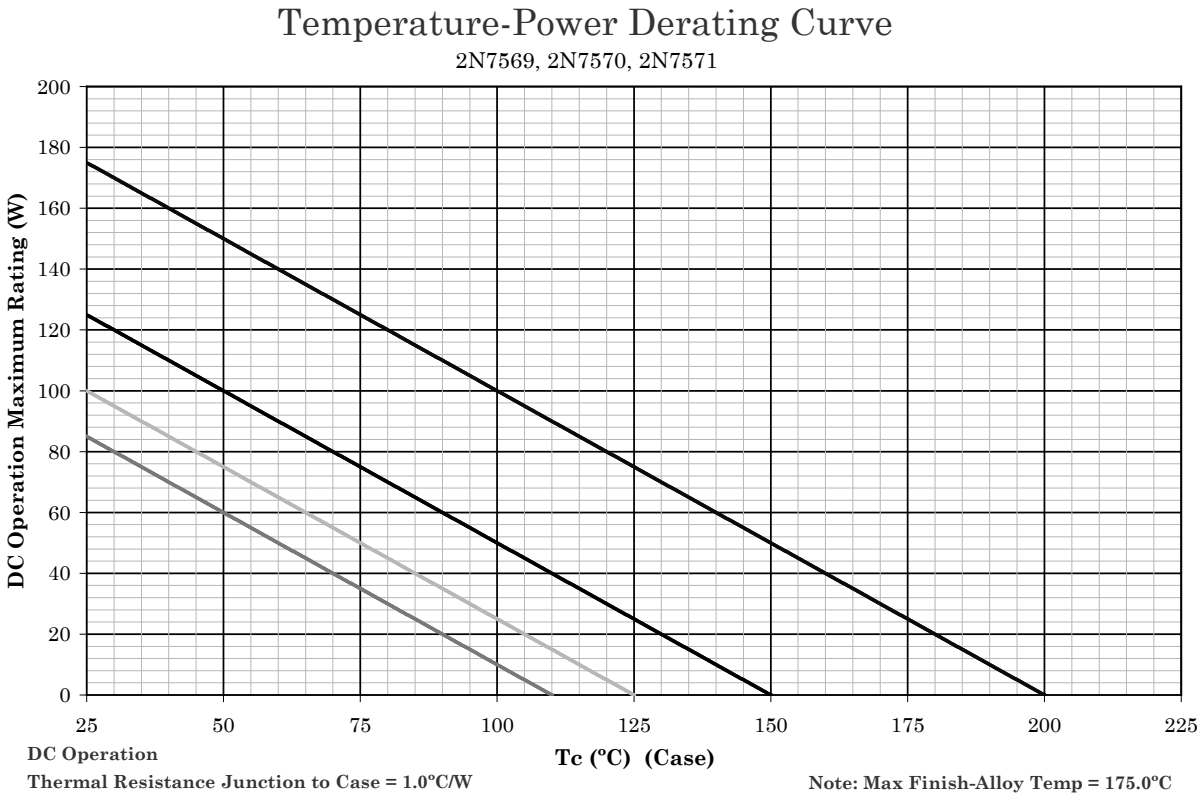
1/ For sampling plan, see MIL–PRF–19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) – for qualification and re-qualification only.

Inspection	MIL–STD–750		Qualification sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 80 percent rated; T = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL–PRF–19500.	
Subgroup 5			15 devices c = 0
Barometric pressure (reduced)	1001	Test condition C. See 1.3.	
Subgroup 6			11 devices
Electrostatic discharge sensitivity classification	1020		
Subgroup 8			45 devices c = 0
Reverse voltage leakage stability	1033	Condition A for all devices.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

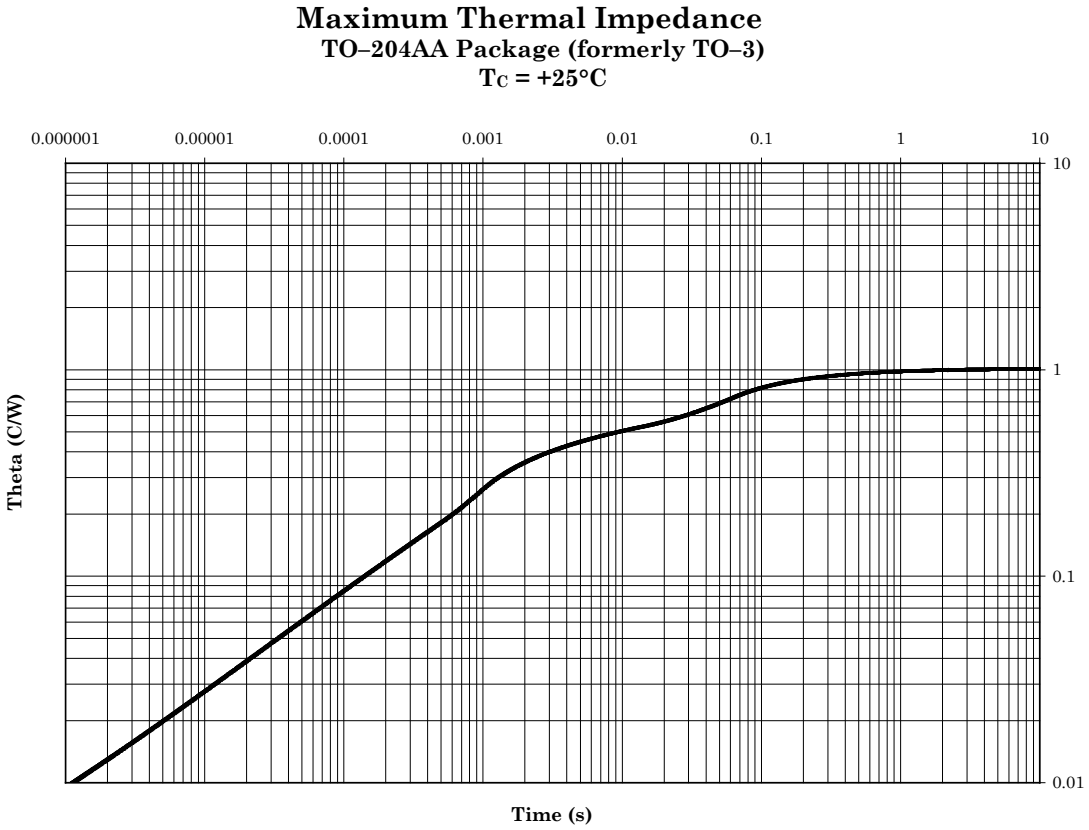


NOTES:

1. All devices are capable of operating at less than or equal to T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 2. Temperature-power derating graph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.



$R_{\theta JC}$ = 1.0 °C/W max.

FIGURE 3. Thermal impedance graph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

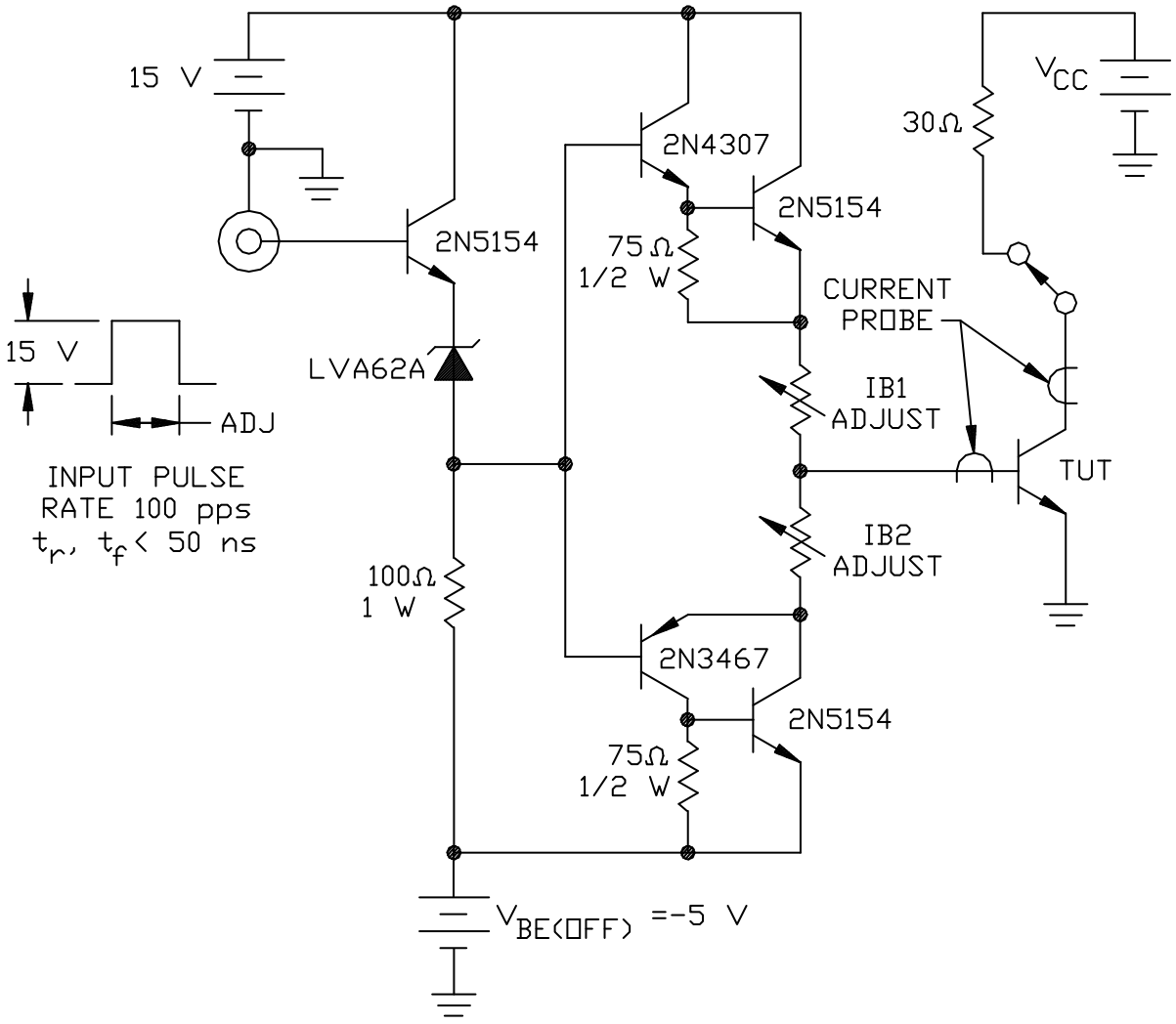


FIGURE 4. Pulse response test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

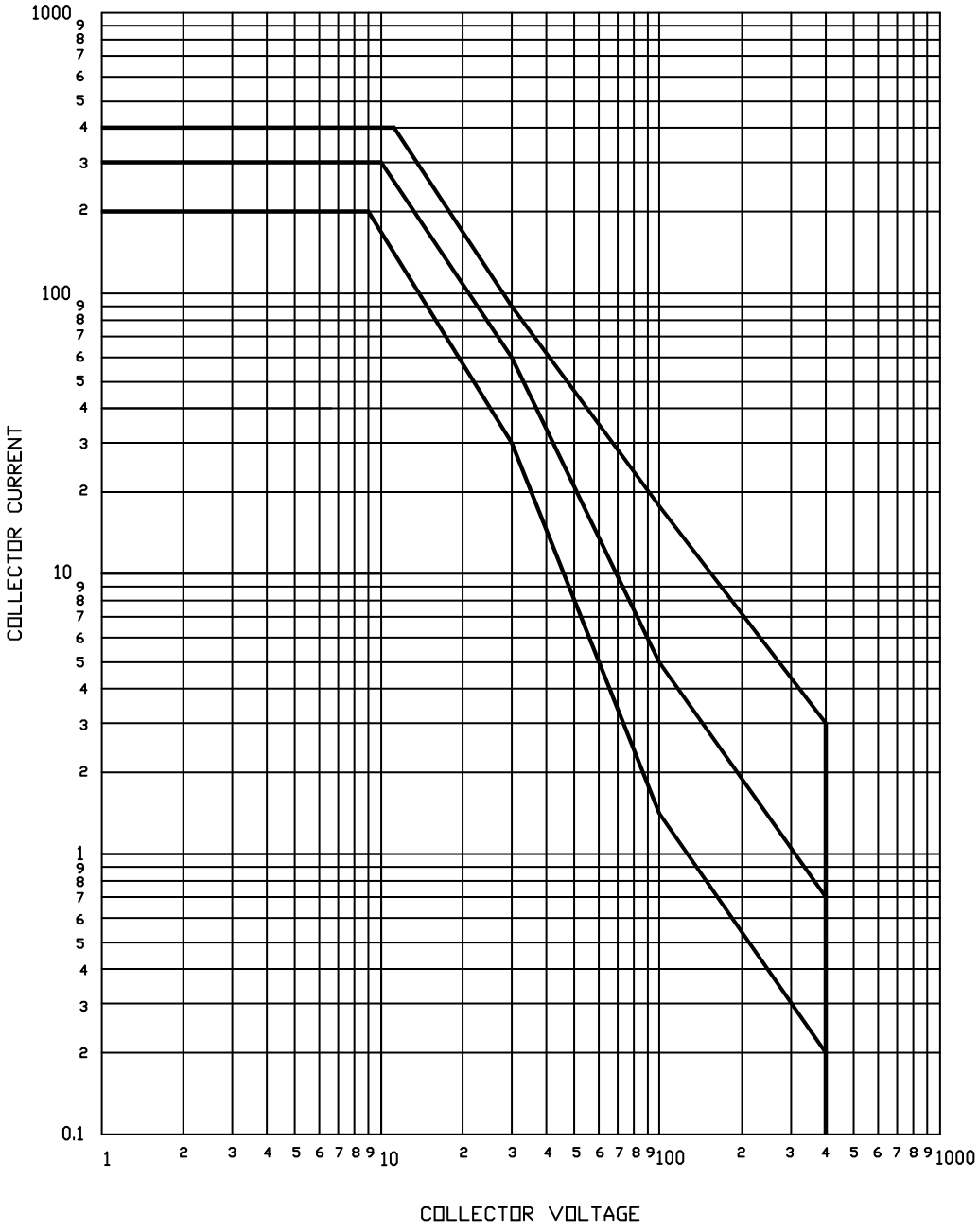


FIGURE 5. Maximum safe operating area graph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL–PRF–19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Quality level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.

Custodians:
Army – CR
Navy – EC
Air Force – 85
NASA – NA
DLA – CC

Review activities:
Army – MI
Air Force – 99

Preparing activity:
DLA – CC

(Project 5961–2012–005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:19Z
Check the source to verify that this is the current version before use.